<table>
<tr><td>Course Title</td><td colspan="6">Database Systems</td></tr>
<tr><td>Course Code</td><td colspan="6">CC-215</td></tr>
<tr><td>Credit Hours</td><td colspan="6">3 (3,0)</td></tr>
<tr><td>Category</td><td colspan="6">Computing Core</td></tr>
<tr><td>Prerequisite</td><td colspan="6">None</td></tr>
<tr><td>Co-Requisite</td><td colspan="6">None</td></tr>
<tr><td>Follow-up</td><td colspan="6">DI-324: Database Administration and Management</td></tr>
<tr><td>Course Introduction</td><td colspan="6">The course aims to introduce basic database concepts, different data models, data storage and retrieval techniques and database design techniques. The course primarily focuses on relational data model and DBMS concepts</td></tr>
<tr><td rowspan="5">Course Learning Outcomes (CLOs)</td><td colspan="4">At the end of the course, the students will be able to:</td><td>BT</td><td>PLO</td></tr>
<tr><td colspan="4">CLO1: Understand and explain the fundamental database concepts</td><td>C2 (Understand)</td><td>1,2</td></tr>
<tr><td colspan="4">CLO2: Analysis and Design of conceptual, logical and physical database schemas using different data models.</td><td>C4 (Analysis)</td><td>2,3,4,5</td></tr>
<tr><td colspan="4">CLO3: To learn SQL and the fundamentals of PL/SQL</td><td>C3 (Apply)</td><td>2,3,4,5</td></tr>
<tr><td colspan="4">CLO4: To be able to design and evaluate a database system for small business organizations</td><td>C6 (Create)</td><td>2,3,4,5,6,7,10</td></tr>
<tr><td>Syllabus</td><td colspan="6">Introduction: Basic database concepts, Database approach vs. file-based system, database architecture, three level schema architecture, data independence Database Models: Conceptual, Logical, and Physical models. Relational Data Model: Basic terminologies, Attributes and their types, Relational Keys, Integrity Constraints. Relational Algebra: Relational operators, selection, projection, Cartesian product, types of joins. Entity Relationship (E-R) Model: Basic Notations, Degree and Cardinality of a relationship, Different types of entities and relationships, Enhanced ER Model. Structured Query Language (SQL): Introduction, SELECT statement, single row and group functions, Analytic functions, Joins and its types, Subqueries, DDL Commands, Views, Indexes, Sequence, DML Commands, DCL Commands. Normalization: Objectives, Need of Normalization, Functional Dependencies, Armstrong Axioms, Normal Forms. Database Design: Top-down vs. Bottom-up approaches, Strategies for improving database performance. Transaction Management and Concurrency Control: Transaction Properties and Types, Database Recovery Techniques, Concurrency problems, Concurrency control methods. Distributed DBMS: Objectives, Components and Architecture of DDMS, Distributed database transparency features. Advancements in Database Systems: NoSQL Systems, Data Quality and Data Integration, Future trends. Procedural SQL: PL/SQL blocks, Operators, if condition, loops, cursor, Stored procedures and functions, Triggers.</td></tr>
<tr><td>Suggested Instructional/ Reading Material</td><td colspan="6">A. Carlos Coronel, Steve Morris: “Database Systems” Design, Implementation, Management, 13th Edition, ISBN: 978-1-337-62790-0
B. Jeffrey A. Hoffer, Ramesh Venkataraman, Heikki Topi: Modern Database Management, 12th Edition, Pearson, 2015. ISBN-10: 0133544613
C. Thomas Connolly, Carolyn Begg: Database Systems: A Practical Approach to Design, Implementation and Management, 6th Edition, Pearson, 2015. ISBN-10: 1292061189
D. Ramez Elmasri, Shamkant B. Navathe: Fundamentals of Database Systems, 7th Edition, Pearson, 2016. ISBN-10: 1292097612
E. Abraham Silberschatz, Henry F. Korth, S. Sudarshan: Database System Concepts, 7th Ed., McGraw Hill, 2019. ISBN-10: 0078022150
F. Catherine M. Ricardo, Susan D. Urban: Databases Illuminated, (3rd Ed., Jones & Bartlett Learning, 2015. ISBN-10:1284056945
G. Jason Price: Oracle Database 11g SQL (Oracle Press), 1st Ed., McGraw Hill, 2007. ISBN-10: 0071498508
H. Nancy Greenburg, Priya Nathan: Introduction to Oracle 9i, Oracle Press.
I. Sham Tickoo, Sunil Raina: Learning Oracle 11g with PL/SQL Approach, Cadcim Technologies, 2014. ISBN-10: 1932709622</td></tr>
<tr><td rowspan="8">Assessment Plan</td><td>Criteria</td><td>Marks</td><td>CLO1</td><td>CLO2</td><td>CLO3</td><td>CLO4</td></tr>
<tr><td>Quizzes</td><td>12</td><td>4</td><td>4</td><td>4</td><td>0</td></tr>
<tr><td>Assignments</td><td>05</td><td>0</td><td>4</td><td>0</td><td>1</td></tr>
<tr><td>Presentation</td><td>03</td><td>1</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>Project</td><td>05</td><td>0</td><td>1</td><td>0</td><td>4</td></tr>
<tr><td>Mid Term</td><td>35</td><td>10</td><td>10</td><td>15</td><td>0</td></tr>
<tr><td>Final Term</td><td>40</td><td>10</td><td>5</td><td>15</td><td>10</td></tr>
<tr><td>Total</td><td>100</td><td>25</td><td>25</td><td>35</td><td>15</td></tr>
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Week No.	Lecture No.	Topic	Source Book-Chapter No. (Sections / Pages)	Recommendations for Learning Activities
1	1	**Introduction** • Data, Information, Database • File Processing System and its Disadvantages • Database Systems and its Advantages • Types of Databases	**A:** Ch1 (1.1 to 1.3, 1.5, 1.6, 1.8) **B:** Ch1 (p2 to p15, p27 to p30) **C:** Ch1 (1.1 to 1.6) **D:** Ch1 (1.1 to 1.3, 1.5 to 1.7) **E:** Sections 1.1, 1.2 **F:** Sections 1.1 to 1.3, 1.5, 2.1, 2.2	Books Readings
	2	**Structured Query Language (SQL):** • Introduction • SELECT statement • Arithmetic operators • Relational and Logical operators	**A:** Ch3 (3.4), Ch7 (7.1 to 7.4) **C:** Ch2 (2.2) **D:** Ch2 (2.3), Ch8 (8.1 to 8.3) **H:** Ch1 (p2 to p15), Ch2 (p28 to p34) **I:** Ch1 (1.1 to 1.13), Ch2 (2.2 to 2.9) **J:** Chapter 1	Lab Exercises
2	3	**Database Architecture and Components** • Three Level Architecture • Data Independence and its types • Components of Database Environment • Database Languages • Functions of DBMS • Types and Roles of Database Users	**A:** Sections 1.7, 3.5 **B:** Ch1 (p15, p16, p22, p23, p24) **C:** Ch2 (2.1 to 2.4) **D:** Sections 1.8, 2.2 **E:** Sections 1.3, 1.6, 1.7, 1.8 **F:** Sections 1.4, 2.4, 2.5	Books Readings Homework Initiation of Term Project
	4	**SELECT Statement** • Special operators (BETWEEN, IN, LIKE, IS NULL) • Use of DISTINCT • Column Alias • ORDER BY clause **SQL Functions:** • Single row functions ○ Character functions • NVL, NVL2, NULLIF	**H:** Ch2 (p35 to p45), Ch2 (p90 to p101) **I:** Ch1 (1.14 to 1.25), Ch2 (2.10 to 2.27), Ch3 (3.2 to 3.11) **J:** Ch3 (p2 to p34)	Lab Exercises
3	5	**Database Development and Data Modeling:** • Database Development Process • Database Life Cycle • Basic Building Blocks of Data Models (Entities, Attributes, Relationship, Constraints) • **Types of attributes**	**A:** Sections 1.4, 2.1 to 2.4, 3.6 **B:** Ch1 (p17 to p22), Ch3 (p53 to p56) **E:** Section 1.5 **F:** Section 2.6	**Quiz#1** Submission of one Page Proposal for Term Project Books Readings
	6	**SQL Functions:** • Single row functions (Character, Number, Date) • Type conversion functions	**H:** Ch4 (p101 to p127) **I:** Ch5 (5.2 to 5.27)	Lab Exercises
4	7	**Logical Data Models** • Hierarchical Data Model • Network Data Model • Relational Data Model ○ Relational Keys • Integrity Constraints / Rules • Entity Integrity, Referential Integrity	**A:** Sections 2.5 to 2.6, 3.1 to 3.3, 3.6 to 3.9 **B:** Ch1 (p24 to p27) **C:** Sections 4.1 to 4.3 **D:** Sections 2.1, 2.3 to 2.5, Chapter 5 **E:** Sections 1.9, 2.1 to 2.4 **F:** Sections 1.6, 2.7, 4.1 to 4.4	Books Readings Homework

	8	**Relational Algebra**
		• Operators of relational algebra
		SQL Functions:
		  ○ Decode function
		  ○ CASE expression
		  ○ Multi-row (group) functions
		• Group by clause
		• Having clause
		• **Analytic functions**
			A: Section 3.4
			C: Section 5.1
			D: Sections 8.1 to 8.5
			E: Sections 2.5, 2.6
			F: Section 4.5
			H: Ch2 (p45 to p51)
			I: Ch4 (4.2 to 4.12)
				Lab Exercises
				Assignment#1

5	9	**Entity Relationship (E-R) Modeling:** • Introduction • Basic Constructs and Notations • Connectivity / Cardinality of a Relationship • Relationship Strength (Existence Dependency) • Relationship Participation • Degree of Relationship	**A:** Ch4 (4.1, 4.2) **B:** Ch2 (p56 to p78) **C:** Ch12 (12.1 to 12.3) **D:** Ch3 (3.1 to 3.3)	**Quiz#2** Books Readings Lab Exercises
	10	**SQL Joins** : Cartesian Join, Inner or Equi Join, Outer Join (Left, Right, Full), Non-Equi Join, Self Join	Oracle Press: Ch4	Submission of Preliminary Report of Term Project

6	11	**Entity Relationship (E-R) Model:** • Modeling Multivalued Attribute(s) • Composite Entities • Developing ER diagram – Examples	**A:** (4.2, 4.3) **B:** Ch2 (p57 to p90) **C:** Ch12 (12.4, 12.5) **D:** Ch3 (3.4, 3.5)	**Lab:** To Practice the SQL Data Management Commands, Complex queries.
	12	**Subqueries**	Oracle Press: Ch4, Ch6, Ch9	Lab Exercises
7	13	**Enhanced Entity Relationship (E-R) Model:** • Entity Super types and subtypes • Comparison of E-R Modeling Symbols	**A:** Ch5 (5.1), **B:** Ch2(p114 to p128), **C:** Ch13 (13.1 to 13.3) **D:** Ch4 (4.1 to 4.3)	Case Studies Tiny College (A) Pine Valley Furniture Company (B)
	14	• Creating Database Tables • Managing Table and Column Level Constraints	Oracle Press: Ch8, Ch10	Submission of Assignemnt#1 Submission of E-R Model of the Project
8	15	**Transform E-R Diagram into Database Structure:** General Rules Governing Relationships Among Tables	**A:** Ch9 (9.6) **B:** Ch4 (p165 to p178)	**Quiz#3**
	16	**Data Manipulation:** Insert, Update and Delete operations **Revision**		
Mid Term Exam				
9	17	**Normalization of Database Tables:** Need for Normalization, Functional dependencies, Armstrong axioms	**A:** Ch6 (6.1 to 6.3) **B:** Ch4(p178 to p184) **C:** Ch14 (14.1 to 14.6) **D:** Ch14 (14.1 to 14.3)	Books Readings
	18	**Database Objects:** Virtual Tables (Views), Indexes, Sequence, Synonyms	Oracle Press: Ch11, Ch12	
10	19	**Normalization:** First, Second and Third Normal Forms	**A:** Ch6 (6.3 to 6.5), **B:** Ch4(p185 to p187) **C:** Ch14 (14.7 to 14.9) **D:** Ch14 (14.4)	Books Readings Submission of Revised E-R Model of Final Project
	20	**Controlling User Rights & Access (DCL):** Role, Privileges, Grant, Revoke	Oracle Press: Ch13	**Assignment#2**
11	21	**Normalization and Database Design:** Higher Level Normal Forms, BCNF, 4NF, 5NF	**A:** Ch6 (6.6, 6.7), **B:** Ch4 (p187 to p192) **C:** Ch15, **D:** Ch14 (14.5 to 14.8)	**Quiz#4**
	22	**Database Design:** • Top-down vs. Bottom-up approach • Phases of database design (Conceptual, Logical, Physical)	**A:** Ch6 (6.8), Chp9 **B:** Ch5 **C:** Ch10	Books Readings
12	23	**Transaction Management:** What is a Transaction; Evaluating Transaction Results, Transaction Properties, Transaction Management with SQL, Buffer Management, Recovery Management	**A:** Ch10 (10.1, 10.7) **C:** Ch22 (22.1, 22.3) **D:** Ch20	Submission of Assignment#2 Presentations Books Readings
	24	**Procedural SQL:** Basics of PL/SQL		Presentations
13	25	**Concurrency Control:** Lost Updates, Un-committed Data, Inconsistent Retrievals, Dirty Data	**A:** Ch10 (10.2, 10.3) **C:** Ch22 (22.2, 22.5) **D:** Ch21	Books Readings **Quiz#5** **Assignment#3**
	26	**PL/SQL:** Selection, Repetition, Exception Handling, Cursors		
14	27	**Concurrency Control:** Locking, Granularity, Lock Types, Two-Phase Locking, Deadlocks, Time Stamping Methods	**A:** Ch10 (10.4, 10.5) **C:** Ch22 (22.2, 22.5) **D:** Ch21	Books Readings
	28	**PL/SQL:** Functions, Stored Procedures		
15	29	**Distributed Databases:** Introduction, Types of distributed databases, Advantages & Disadvantages, Components of DDBMS	**A:** Ch12 **C:** Ch24 (24.1 to 24.5) **D:** Ch23 (23.1,23.2)	Submission of the project
	30	**PL/SQL:** Database Triggers		**Quiz#6**, Submission of Assignment#3
16	31	Big Data and NoSQL • Data Quality and Data Integration	**A:** Ch14 (14.1 to 14.5) **B:** Ch 10	Books Readings
	32	Future Trends, Revision		Preparation for the Final Exam